Question			Answer	Marks	AO	Guidance	
13			$\int_1^n x^2\,\mathrm{d}x + \frac{n^2+1^2}{2} + \frac{1^2-2^2}{12} - \frac{n^2-(n+1)^2}{12}$	B1	1.1a	Substitute correctly into formula Condone r but not n instead of x	
			$\left[\frac{x^3}{3}\right]_1^n + \frac{6n^2+6+1-4-n^2+n^2+2n+1}{12}$				
			$\frac{n^3-1}{3} + \left[\frac{6n^2+2n+4}{12}\right]$	B1	1.1	Integral correctly evaluated	
			$\frac{2n^3+3n^2+n}{6}$	B1	2.1	Correct single fraction – may not be fully simplified	OR $\frac{n(n+1)(2n+1)}{6}$ $= \frac{2n^3+3n^2+n}{6}$
			$\frac{n(2n^2+3n+1)}{6} = \frac{n(n+1)(2n+1)}{6}$	B1	2.2a	Correct completion Intermediate step needed Dep on **B3**	OR convincing comparison of $\frac{n(n+1)(2n+1)}{6}$ and $\frac{n^3-1}{3} + \frac{6n^2+2n+4}{12}$
				[4]			